

ICMP Guide for Thaghrach

Description

ICMP (Internet Control Message Protocol) is a supporting protocol in the Internet Protocol Suite. It operates at the **Network Layer** and is primarily used for **error reporting**, **diagnostics**, and **network troubleshooting**.

Unlike TCP or UDP, ICMP is not used to transfer application data. Instead, it helps devices communicate information about network problems, such as when a packet cannot be delivered, or to test whether a remote host is reachable. ICMP messages are encapsulated directly within IP packets.

The most well-known uses of ICMP are the **Ping** utility (Echo Request/Reply) and **Traceroute**.

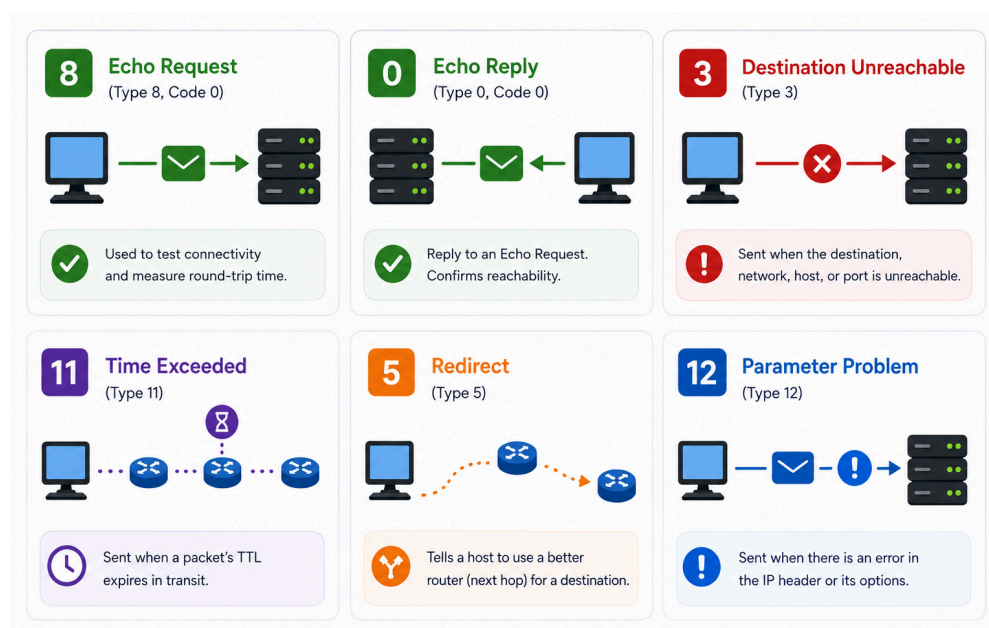
Key Components of ICMP

1. ICMP Message Structure

Every ICMP message includes:

- **Type** (8 bits) — Defines the main purpose of the message
- **Code** (8 bits) — Provides additional detail about the Type
- **Checksum** — For error detection
- **Data** (variable) — Specific to the message type

2. Most Important ICMP Types

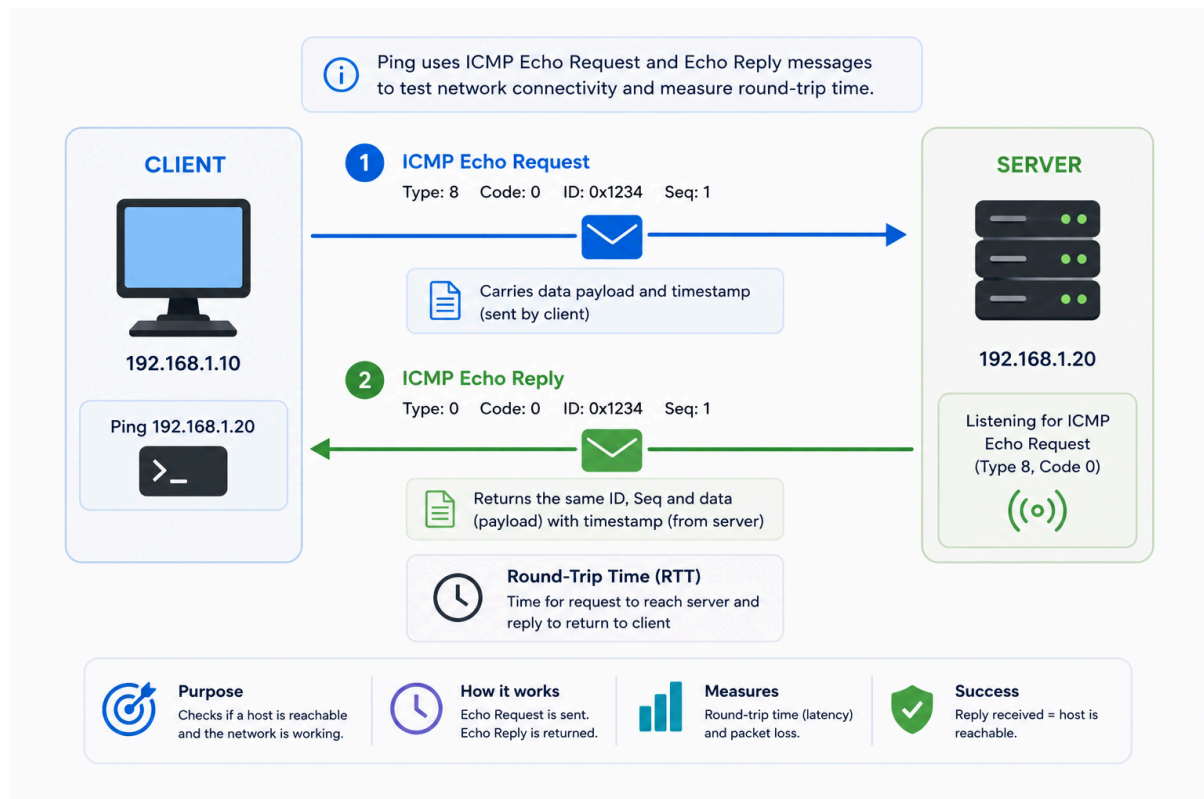


3. Common Use Cases

- Testing reachability (Ping)
- Discovering network paths (Traceroute)
- Reporting delivery errors
- Informing about congestion or routing issues

Visualizations

ICMP Echo Request / Reply (Ping):



What is ICMP?

<https://youtu.be/xTqtm7-k25o?si=WRNCvhBKNhXleY1S>